ODE - 1

Date →

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Define a differential equation.

Any equation which involves derivatives of one or more than one dependent variables w.r.t one or more than one independent variables

Ordinary differential equation

A differential eqn. which involves derivatives of one or more than one dependent variables w.r.t only one independent variable is called an ODE.

Eg: ① $\frac{d^2y}{dx^2} + 2y = \sin x$ ② $\frac{dy}{x} + 2\frac{dz}{dx} = e^x$

There is only one independent variable

ORDER of an ODE

The highest order derivative occuring in the ODE is called the order of the ODE. In ① order is 2, in ② order is 1.

Degree of an ODE

The degree of a differential eqn is the exponent of the higher order derivative occuring in that ODE after making the ODE free from radicals, fractions, transcendental fn's. as far as derivatives are concerned.

Examples ① $\left(\frac{d^2y}{dx^2}\right)^{2/3} = \frac{y^{1/2}}{x^{1/3}(x^{2/3}+1)}$

$\Rightarrow \left(\frac{d^2y}{dx^2}\right)^2 = \frac{y^{3/2}}{x(x^{2/3}+1)^3}$ → degree 2, order 2

② $\sin\left(\frac{dy}{dx}\right) = x \Rightarrow \sin^{-1}x = \frac{dy}{dx}$ → order 1, degree 1 iff $0 \le \frac{dy}{dx} \le \frac{\pi}{2}$

⊛ If $\frac{dy}{dx}$ can take any value, then degree will not exist but order is 1.

③ $\frac{dy}{dx} + \int y\,dx = 0$ → not a DE (but integro - DE)

$\Rightarrow \frac{d^2y}{dx^2} + y = 0$ → order 2, degree 1

Linear / Nonlinear ODE

A DE is said to be linear if

① every dependent variable and every derivative occuring in the ODE should occur in first degree only

② No product of dependent variable and/or derivatives should be there.

example ① $\frac{dy}{} + \frac{dz}{} = \boxed{yz}$ → product term ⇒ nonlinear be there.

Example

① $\frac{dy}{dx} + \frac{dz}{dx} = \boxed{yz}$ → product term → nonlinear

② $\frac{d^2y}{dx^2} + y^2 = \sin x$ → not linear due to product of dependent variable

③ $y \cdot \frac{dy}{dx} + \sin x = y$ → nonlinear

④ $\frac{dy}{dx} + \underline{f(x)} \cdot y = \underline{g(x)}$ → linear

Formation of ODE from given equations

① Find the differential eqn. for the family of straight lines

$y = mx + c \Rightarrow \boxed{\frac{dy}{dx} = m}$ $\boxed{\frac{d^2y}{dx^2} = 0}$

② Find the DE for the family of curves $y = Ae^{Bx+c}$

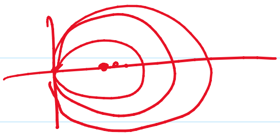
$\frac{dy}{dx} = ABe^{Bx+c}$ $\Rightarrow B = \frac{d^2y}{dx^2} \Big/ \frac{dy}{dx}$

$\frac{d^2y}{dx^2} = AB^2 e^{Bx+c}$

also, $B = {}^{dy}/_{dx} \Big/ y$

$\therefore \boxed{\frac{d^2y}{dx^2} \cdot y = \left(\frac{dy}{dx}\right)^2}$

③ Find the DE for family of circles passing through origin and center at x-axis.



$(x-a)^2 + y^2 = a^2$ (eqn. of circles)

$\therefore x^2 - 2ax + y^2 = 0$

$2x - 2a + 2y \cdot {}^{dy}/_{dx} = 0$

$2 + 2\left(\frac{dy}{dx}\right)^2 + 2y \cdot \frac{d^2y}{dx^2} = 0$

$\therefore y \cdot \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 1 = 0$

④ Find DE for $y = A\cos mx + B\sin mx$ where A, B are arbit. consti.

${}^{dy}/_{dx} = -Am\sin mx + Bm \cdot \cos mx$

$\frac{d^2y}{dx^2} = -Am^2\cos mx - Bm^2\sin mx$

$= -m^2 y$

$\boxed{\frac{d^2y}{dx^2} + m^2 y = 0}$

$$\boxed{\frac{d^2y}{dx^2} + m^2 y = 0}$$

Homogeneous and Non-homogeneous ODE

$dy/dx + P(x)\, y = Q(x) \to$ Non-homogeneous

$\qquad\qquad\qquad\quad$ └ $0 \to$ homogeneous

Eg: **Relation between acceleration and velocity**

(*) acceleration → e^{-t} m/sec²

Initial velocity = 0

$\frac{dv}{dt} = e^{-t} \Rightarrow dv = e^{-t}\; vt$

$\Rightarrow v = -e^{-t} + c$

$t = 0 \Rightarrow v = 0 \Rightarrow c = 1$

$\Rightarrow \boxed{v = 1 - e^{-t}}$

(*) $\boxed{dy/dt = y} \Rightarrow dy/y = dt \Rightarrow \ln y = t + c$

$\Rightarrow y = c_1 e^{t}$ (general soln.)

↑

This can be soln. to many functions,

$y = e^{t}$
$y = 0$
$y = 3e^{t}$ } and many more

So, the $\boxed{\text{general soln}}$ is what we seek

→ giving one soln. will not help.

→ So, general soln. is $\boxed{y = c_1 e^{t}}$ (c_1 arbit const.)

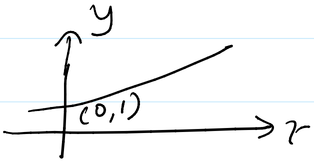
(*) $dy/dt = y$, $y(0) = 1$ (IC)

$dy/y = dt \Rightarrow \ln y = c_1 + t$

$\Rightarrow \ln 1 = c_1 + 0$

$\Rightarrow c_1 = 0$

$\Rightarrow y = e^{t}$



(*) $\frac{dy}{dt} = 2y$, $y(0) = 1$ (homogeneous DE)

dt

$$\int \frac{dy}{2y} = \int dt \quad \text{or} \quad \int \frac{dy}{y} = \int 2\,dt$$

Integrating $\Rightarrow \ln y = 2t + c_1$

$$\Rightarrow \quad y(t) = c \cdot e^{2t}$$

$$\Rightarrow \quad y(0) = 1 \quad \Rightarrow c = 1$$

$$\Rightarrow \quad \boxed{y(t) = e^{2t}}$$